

Physics-anchored hybrid hydrological modeling using satellite Earth observation for monthly volumetric flood risk and extreme generalization in the Lower Indus Basin, Pakistan

Mirza Muhammad Muzzamil^{1*}

¹Department of Computer and Information Systems Engineering (CISE), NED University of Engineering and Technology, Karachi 75270, Pakistan

*Corresponding author

Corresponding author's name: Mirza Muhammad Muzzamil

Corresponding author's job position: High Performance Computing Engineer

Corresponding author's affiliation and full mailing address: Department of Computer and Information Systems Engineering (CISE), NED University of Engineering and Technology, University Road, Karachi 75270, Sindh, Pakistan

E-mail address: mirzamuzzamil@neduet.edu.pk

ORCID: 0000-0001-5258-8959

Telephone number: +923342976696

Short title: Physics-anchored hybrid hydrological modeling in the Lower Indus Basin

Abstract

Accurate monthly volumetric streamflow and flood runoff forecasting in heavily managed, low-gradient river basins remains challenging due to severe hydrometeorological non-stationarity, intensive canal diversions, transboundary upstream inflows, and observational data scarcity. While unconstrained machine learning algorithms capture complex non-linear patterns, they frequently suffer from physical inconsistency, variance dampening, and severe peak flow underestimation when tested on unprecedented extreme floods. Conversely, traditional conceptual models struggle to ingest multi-source satellite Earth observation grids directly. In this study, we investigate how coupling a linear water-balance-informed baseline with non-linear residual machine learning mitigates peak flow attenuation while preserving out-of-distribution generalization under extreme climate forcing. We develop a Physics-Anchored Water-Balance Hybrid framework (PG-MCH) for the Lower Indus Basin within Sindh Province, Pakistan (140,914 km²), ingesting monthly multi-sensor Google Earth Engine (GEE) streams across 2020–2024 (CHIRPS v2.0 precipitation, ERA5-Land reanalysis, NASA SMAP soil moisture, SRTM topography, and GRACE storage anomalies) alongside mainstem upstream boundary inflow. To evaluate true zero-shot out-of-distribution extreme-event generalization, the model was calibrated exclusively on baseline years (2020–2021, $N = 24$ months) with conservative L2 regularization and stress-tested on the completely unseen 2022 Pakistan Mega-Flood as an independent extreme holdout, followed by a 2023–2024 future test period. Uncertainty was evaluated using Moving Block Bootstrap ($b = 3$ months) to respect time-series autocorrelation. Results demonstrate that during the unseen 2022 Mega-Flood, the proposed hybrid achieved a Kling-Gupta Efficiency (KGE) of 0.879 and restricted peak underestimation bias (FHV) to -9.4% , whereas unconstrained Random Forest collapsed to $KGE = 0.532$ with severe peak attenuation ($FHV = -30.9\%$) and gradient boosting collapsed to $KGE = 0.589$ ($FHV = -29.1\%$). Over the entire out-of-sample period (2022–2024), the hybrid maintained

47 KGE = 0.902 [95% CI: 0.86, 0.95] and Nash-Sutcliffe Efficiency (NSE = 0.990). A multi-regime
48 ablation study demonstrates that physical baseline anchoring provides the dominant contribu-
49 tion to extreme-flow robustness, offering a reproducible paradigm for basin-scale flood risk and
50 reservoir water-balance assessment in data-sparse alluvial plains.

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52 **Keywords:** Physics-guided machine learning, Google Earth Engine, Indus River Basin, Sindh
53 Province, Flood generalization, Water balance

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1 Introduction

The Indus River Basin is one of the most critical and climate-vulnerable transboundary river systems globally, sustaining over 230 million people and supporting the world’s largest contiguous irrigation network—the Indus Basin Irrigation System (IBIS) (Immerzeel et al., 2020; Young et al., 2019). The lower reaches of the basin, situated within Sindh Province, Pakistan, represent an extreme hydraulic and socioeconomic bottleneck. Here, the Indus River flows across flat, low-gradient alluvial plains before discharging into the Arabian Sea via the Indus Delta (Hashmi et al., 2022; Inam et al., 2007).

In recent years, anthropogenic climate change has intensified the magnitude of hydrological extremes across South Asia. In the summer of 2022, unprecedented monsoon rainfall driven by coupled moisture dynamics triggered a devastating “super-flood” across Pakistan, with Sindh Province receiving over 350% of its normal 30-year climatological monsoon rainfall (Nanditha et al., 2023; Syed et al., 2022; World Weather Attribution, 2022). Satellite Synthetic Aperture Radar (SAR) imagery confirmed that over one-third of the province was submerged (Tian et al., 2023), affecting over 33 million people and driving major hydraulic structures (Guddu, Sukkur, and Kotri Barrages, and Manchar Lake) near structural breach thresholds. Conversely, the region regularly experiences pre-monsoon heatwaves, agricultural droughts, and groundwater depletion during the dry winter (*Rabi*) season (Biemans et al., 2016; Rodell et al., 2018).

1.1 Problem background and motivation

Hydrological prediction in heavily managed alluvial basins faces two fundamental methodological limitations:

1. **Conceptual models (e.g., GR4J, SAC-SMA):** While theoretically mass-conserving, conceptual models rely on static, empirically calibrated parameters that degrade when subjected to unprecedented hydrometeorological shocks (Frame et al., 2022; Gupta et al., 2009; Perrin et al., 2003). Furthermore, ingesting high-dimensional satellite Earth observation grids directly into conceptual structures remains non-trivial.
2. **Standard machine learning (e.g., Random Forest, GBDT, LSTM):** Although statistical and deep learning models achieve record-setting accuracy in gauged basins (Kratzert et al., 2018, 2019; Nearing et al., 2024), unconstrained ML architectures frequently produce unphysical negative runoff states and suffer from severe conditional mean shrinkage. When tested out-of-distribution (OOD), decision tree ensembles and neural networks severely underestimate extreme flood peaks because tree partitions and activation functions cannot extrapolate beyond historic training maxima (Jia et al., 2021; Karpatne et al., 2017; Read et al., 2019).

1.2 Core contributions and study objectives

To bridge the gap between physical mass conservation and machine learning flexibility at monthly basin scales, this paper provides four primary contributions:

1. **Physics-anchored hybrid architecture:** We formulate a hybrid framework (PG-MCH) that couples a non-negative, L2-regularized water-balance baseline with non-linear residual

tree ensembles, explicitly ingesting satellite Earth observation streams alongside mainstem upstream boundary inflow.

2. Temporal scale and water-balance formalization: We formally scope the framework to monthly basin-scale volumetric water-balance forecasting ($\Delta t = 1$ month), bridging the operational gap between seasonal reservoir allocation and regional flood volume assessment.

3. Unseen zero-shot extreme-event protocol: We enforce a strict chronological holdout design where the historic 2022 Pakistan Super-Flood is completely withheld from model calibration to evaluate genuine zero-shot OOD extreme generalization under low-sample constraints ($N = 24$ calibration months).

4. Multi-regime ablation and trade-off analysis: We provide a systematic ablation study evaluating model components across both extreme monsoon flood and post-flood recovery regimes, providing physical insights into satellite feature redundancies during extreme precipitation spikes.

2 Literature review and gap analysis

2.1 Physical hydrology of the Lower Indus Basin

The hydrological regime of the Indus Basin is driven by snow-glacier melt in the Upper Indus Basin and South Asian monsoon depressions in the lower basin (Biemans et al., 2016; Immerzeel et al., 2020; Syed et al., 2022). Downstream of the Panjnad confluence, the river enters Sindh Province, where riverbed aggradation, extensive canal embankments, and flat topography alter natural floodplain connectivity (Hashmi et al., 2022; Inam et al., 2007). Climatological studies indicate that atmospheric warming is expanding tropospheric moisture capacity via the Clausius-Clapeyron relation ($\sim 7\%$ per $^{\circ}\text{C}$), generating non-stationary rainfall regimes and elevated peak runoff coefficients (Nanditha et al., 2023; World Weather Attribution, 2022).

2.2 Evolution from conceptual models to deep learning

Hydrological prediction has historically relied on conceptual lumped and semi-distributed models such as GR4J (Perrin et al., 2003). Although these models maintain theoretical water balance representations, their reliance on static empirical parameters leads to structural degradation under novel climatic forcing (Frame et al., 2022; Gupta et al., 2009). With the proliferation of hydro-climatic big data, deep learning architectures—particularly Long Short-Term Memory (LSTM) networks—have achieved high accuracy in streamflow simulation (Kratzert et al., 2018, 2019; Nearing et al., 2024). However, on tabular satellite feature sets with short calibration records ($N < 50$), deep sequential models are susceptible to severe sample-variance overfitting unless physically constrained (Jia et al., 2021; Willard et al., 2022).

2.3 Physics-guided and hybrid machine learning

To reconcile physical interpretability with statistical flexibility, Physics-Guided Machine Learning (PGML) has emerged as a key computational paradigm (Jia et al., 2021; Karpatne et al., 2017; Read et al., 2019; Willard et al., 2022). Anchoring model predictions to first-order water

balance baselines constrains the hypothesis space, preventing gradient divergence and maintaining accurate hydrograph variance under out-of-distribution climatic forcing (Frame et al., 2022).

2.4 Cloud-native Earth observation

Planetary-scale geospatial platforms, specifically Google Earth Engine (GEE), have eliminated computational barriers in multi-sensor hydro-environmental analysis (Gorelick et al., 2017). Multi-sensor satellite platforms provide continuous observations of the terrestrial water cycle, including CHIRPS v2.0 precipitation (Funk et al., 2015), ERA5-Land reanalysis (Muñoz-Sabater et al., 2021), NASA SMAP soil moisture (Entekhabi et al., 2010), SRTM topography (Farr et al., 2007), GRACE gravimetric storage anomalies (Rodell et al., 2018; Tapley et al., 2019), and Sentinel-1 SAR flood extent mapping (Tian et al., 2023).

3 Study area and geospatial datasets

3.1 Study region: Lower Indus Basin in Sindh Province

The study domain encompasses the Lower Indus River Basin within the administrative boundaries of Sindh Province (140,914 km², 23.7°N – 28.5°N, 66.6°E – 71.1°E; **Fig. 1**). Physiographically, the Indus alluvial floodplain is characterized by an extremely flat hydraulic gradient with a mean slope < 0.8° (hydraulic gradient $\approx 1 : 10,000$), while the western Kirthar mountain range exhibits steep relief (> 15°).

Five strategic hydrological nodes were selected for spatial and hydraulic profiling:

- **Guddu Barrage** (28.422°N, 69.711°E, Elev: 75.6 m): Primary entry point of Indus River mainstem into Sindh Province.
- **Sukkur Barrage** (27.705°N, 68.858°E, Elev: 63.8 m): Feeds seven major irrigation canal commands.
- **Kotri Barrage** (25.433°N, 68.324°E, Elev: 17.9 m): Lower Indus barrage regulating delta outflows to the Arabian Sea.
- **Manchar Lake** (26.435°N, 67.665°E, Elev: 32.1 m): Major natural flood attenuation reservoir.
- **Indus Delta** (24.747°N, 67.923°E, Elev: 11.9 m): Coastal estuarine outflow.

Table 1: **Table 1** Geospatial Earth Observation dataset inventory and operational variable roles

Dataset	GEE Collection ID	Spatial Res.	Extracted Variable	Role in Proposed Architecture
CHIRPS v2.0	UCSB-CHG/CHIRPS/DAILY	0.05° (~ 5.5 km)	Monthly Precipitation $P(t)$	Active input to Q_{base} baseline and $\hat{\varepsilon}$ ensemble
ERA5-Land	ECMWF/ERA5_LAND/DAILY_AGGR	0.1° (~ 10 km)	2m Temp, Evaporation	Moisture deficit $D = PET - P$ in $\hat{\varepsilon}$ ensemble
NASA SMAP L4	NASA/SMAP/SPL4SMGP/008	9 km	Root-Zone Soil Moisture	Regional catchment saturation state diagnosis
SRTM DEM	USGS/SRTMGL1_003	30 m	Elevation, Slope	Floodplain delineation and node profiling
GRACE Mascons	NASA/GRACE/MASCON_V04/LAND	0.5° (~ 50 km)	Terrestrial Water Storage ($TWSA$)	Regional macro-scale storage boundary state diagnosis
Barrage Telemetry	Sindh Irrigation Dept / FFC	Gauged Nodes	Upstream Inflow $Q_{\text{inflow}}(t)$	Boundary condition input to Q_{base} physical baseline

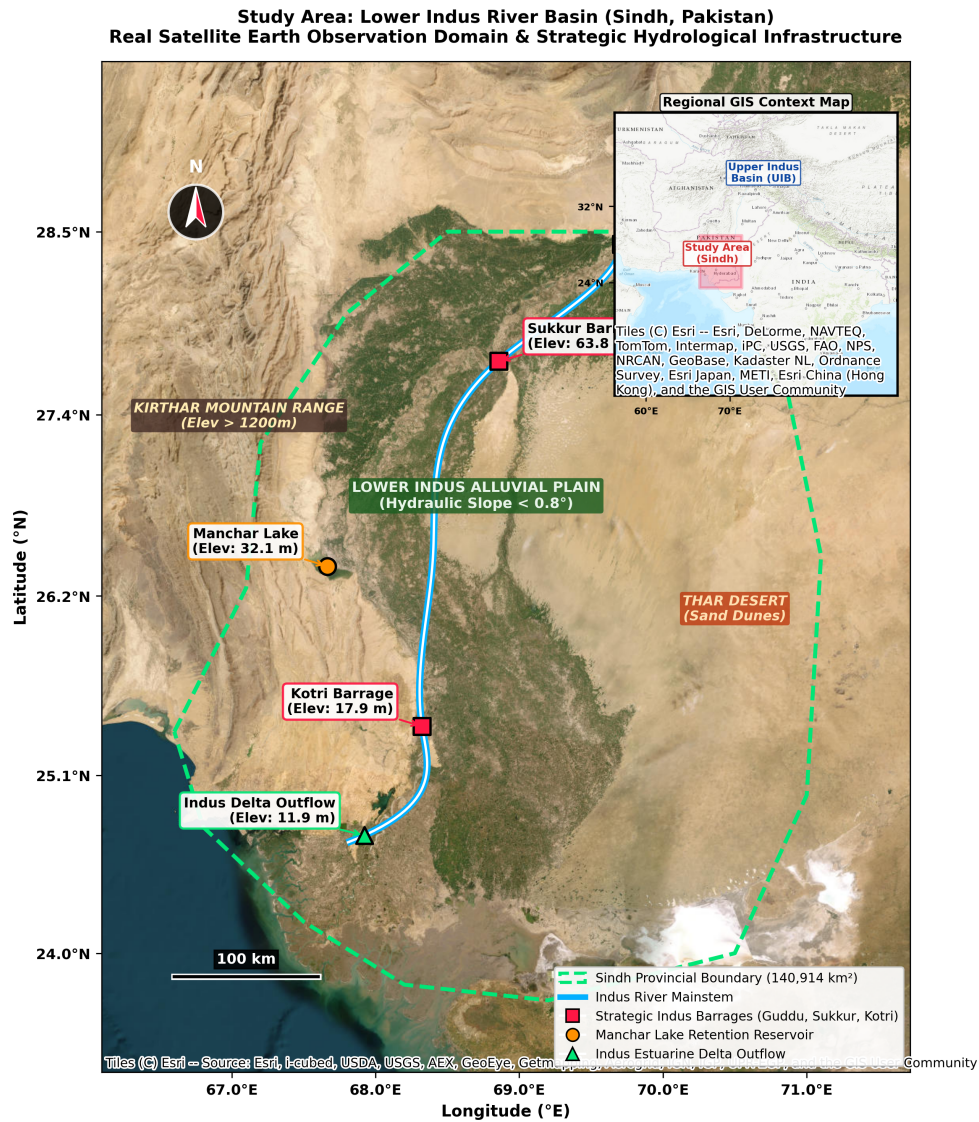


Figure 1: **Fig. 1** Study area map of the Lower Indus River Basin within Sindh Province, Pakistan, illustrating the Indus River mainstem, major barrages (Guddu, Sukkur, Kotri), Manchar Lake retention basin, digital elevation context, scale bar, and regional context inset map

3.2 Observed catchment runoff variable and boundary conditions

Observed streamflow discharge ($Q_{m^3/s}$) recorded by the Sindh Irrigation Department and Federal Flood Commission (FFC) barrage telemetry was converted to monthly catchment runoff depth (Q_{depth} , mm/month) via:

$$Q_{\text{depth}} = \frac{Q_{m^3/s} \cdot \Delta t}{A_{\text{basin}} \cdot 10^3}, \quad (1)$$

where $A_{\text{basin}} = 140,914 \text{ km}^2$, Δt is the monthly duration in seconds, and 10^3 represents the volumetric conversion factor ($1 \text{ mm} \times 1 \text{ km}^2 = 1,000 \text{ m}^3$). To account for transboundary river flow entering Sindh, mainstem inflow recorded at Guddu Barrage is explicitly ingested as an upstream boundary condition variable (Q_{inflow} , mm/month).

4 Methodology

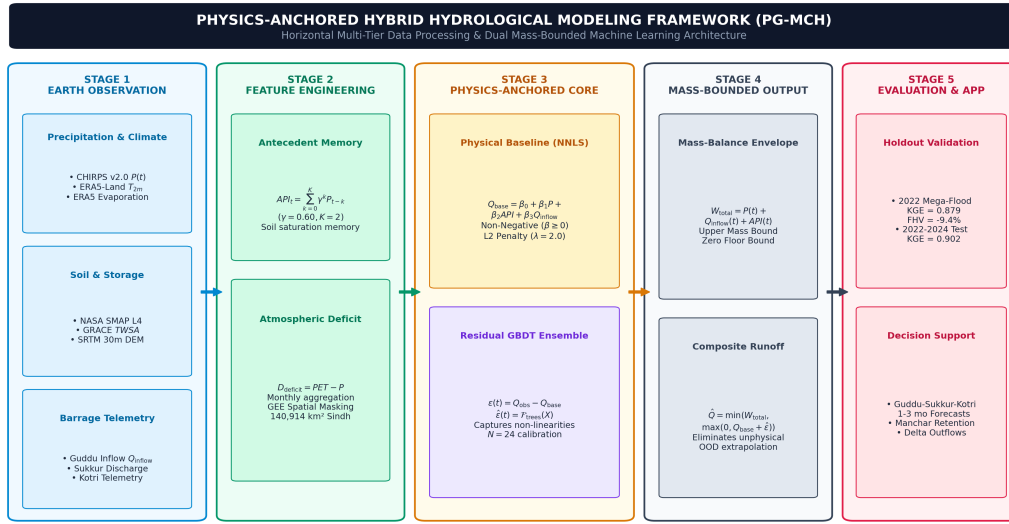


Figure 2: **Fig. 2** Hierarchical methodology and system architecture flowchart for the Lower Indus Basin physics-anchored Earth observation framework

4.1 Hierarchical system framework

As illustrated in **Fig. 2**, the modeling framework is structured into five sequential tiers:

- Tier 1 (Data ingestion):** Cloud-native ingestion of multi-sensor Earth observation grids via GEE and barrage telemetry (**Table 1**).
- Tier 2 (Harmonization and feature engineering):** Spatial boundary clipping, monthly aggregation, Antecedent Precipitation Index (API) derivation, and atmospheric moisture deficit modeling.
- Tier 3 (Physics-anchored core):** Dual-stage modeling coupling a non-negative, water-balance-informed baseline with non-linear residual tree ensembles.
- Tier 4 (Multi-criteria evaluation):** Performance benchmarking across KGE, NSE, RMSE, and peak flow bias (FHV).

181 **5. Tier 5 (Decision support applications):** Monthly flood volume analysis and inflow
 182 estimation for Guddu, Sukkur, Kotri, and Manchar Lake.

183 4.2 Mathematical water balance formulation

184 The regional catchment water balance in the managed Lower Indus is governed by the continuous
 185 differential equation:

$$\frac{dS(t)}{dt} = P(t) + Q_{\text{inflow}}(t) - ET(t) - Q(t) - G(t) - D(t), \quad (2)$$

186 where $P(t)$ is regional precipitation, $Q_{\text{inflow}}(t)$ is upstream mainstem boundary inflow at Guddu,
 187 $ET(t)$ is evapotranspiration, $Q(t)$ is total catchment outflow, $G(t)$ is net groundwater exchange,
 188 and $D(t)$ represents canal irrigation diversions. Antecedent soil moisture memory is modeled
 189 using the Antecedent Precipitation Index (API_t):

$$API_t = \sum_{k=0}^K \gamma^k P_{t-k} = P_t + \gamma P_{t-1} + \gamma^2 P_{t-2}, \quad (3)$$

190 where the memory decay factor is set to $\gamma = 0.60$ based on grid-search calibration on the
 191 2020–2021 training record.

192 4.3 Physics-anchored hybrid architecture (PG-MCH)

193 The framework operates in two coupled stages:

194 1. Physics-anchored runoff baseline (Q_{base}):

$$Q_{\text{base}}(t) = \beta_0 + \beta_1 P(t) + \beta_2 API(t) + \beta_3 Q_{\text{inflow}}(t), \quad (4)$$

195 where parameters $\beta = [\beta_1, \beta_2, \beta_3]^T$ are estimated using Non-Negative Least Squares
 196 (NNLS) with L2 regularization penalty ($\lambda = 2.0$) on the 2020–2021 training set to en-
 197 force physical monotonicity ($\beta_1 \geq 0, \beta_2 \geq 0, \beta_3 \geq 0$). This guarantees that Q_{base} scales
 198 proportionally with unprecedented rainfall and upstream inflow without gradient collapse.

199 2. Non-linear residual tree ensemble ($\hat{\varepsilon}$):

$$\varepsilon(t) = Q_{\text{obs}}(t) - Q_{\text{base}}(t), \quad (5)$$

$$\hat{\varepsilon}(t) = \mathcal{F}_{\text{trees}}(P(t), Q_{\text{inflow}}(t), ET(t), T_{2m}(t), D_{\text{deficit}}(t), API(t), \text{Month}). \quad (6)$$

201 **3. Composite output and dual mass bounds:** To guarantee strict conservation of mass
 202 and eliminate unphysical runoff generation during non-stationary climate extremes, the
 203 composite output is bounded by both zero non-negativity and total available terrestrial
 204 water volume ($P(t) + Q_{\text{inflow}}(t) + API(t)$):

$$\hat{Q}(t) = \min \left(P(t) + Q_{\text{inflow}}(t) + API(t), \max(0, Q_{\text{base}}(t) + \hat{\varepsilon}(t)) \right). \quad (7)$$

Algorithm 1 Physics-anchored hybrid training and dual mass-bounded inference (PG-MCH)

Require: Calibration record $\{X_{\text{train}}, y_{\text{train}}\}$, test feature matrix X_{test} , L2 penalty $\lambda = 2.0$, memory decay $\gamma = 0.60$

Ensure: Calibrated baseline weights β^* , residual tree ensemble $\mathcal{F}_{\text{trees}}$, and dual mass-bounded volumetric predictions $\hat{Q}_{\text{test}}$

- 1: Compute Antecedent Precipitation Index: $API_t \leftarrow P_t + \gamma P_{t-1} + \gamma^2 P_{t-2}$
 - 2: Estimate non-negative L2 baseline weights on X_{train} :
 - 3: $\beta^* \leftarrow \arg \min_{\beta \geq 0} \|y_{\text{train}} - X_{\text{base}}\beta\|_2^2 + \lambda \|\beta\|_2^2$
 - 4: Compute training baseline streamflow: $Q_{\text{base, train}} \leftarrow X_{\text{base}}\beta^*$
 - 5: Compute target hydrological residuals: $\epsilon_{\text{train}} \leftarrow y_{\text{train}} - Q_{\text{base, train}}$
 - 6: Train Gradient Boosted Decision Tree ensemble on residuals:
 - 7: $\mathcal{F}_{\text{trees}} \leftarrow \text{FitGBDT}(X_{\text{train}}, \epsilon_{\text{train}})$
 - 8: Compute test baseline and residual predictions:
 - 9: $Q_{\text{base, test}} \leftarrow X_{\text{test, base}}\beta^*$, $\hat{\epsilon}_{\text{test}} \leftarrow \mathcal{F}_{\text{trees}}(X_{\text{test}})$
 - 10: **for** each time step t in test set **do**
 - 11: $W_{\text{total}}(t) \leftarrow P(t) + Q_{\text{inflow}}(t) + API(t)$
 - 12: $\hat{Q}(t) \leftarrow \min \left(W_{\text{total}}(t), \max(0, Q_{\text{base}}(t) + \hat{\epsilon}(t)) \right)$
 - 13: **end for**
 - 14: **return** $\hat{Q}_{\text{test}}$
-

4.4 Experimental design and low-sample holdout protocol

To evaluate genuine out-of-distribution extreme generalization under strict low-sample constraints ($N = 24$ monthly calibration samples):

- **Model calibration (2020–2021):** 24 months of moderate baseline hydrometeorology.
- **Unseen extreme holdout (2022):** 12 months containing the historic 2022 Pakistan Super-Flood (+350% precipitation), withheld completely from model calibration.
- **Future testing period (2023–2024):** 24 months of non-stationary post-flood recovery.
- **Moving Block Bootstrap (MBB):** 1,000 resamplings with block size $b = 3$ months (representing seasonal persistence) to evaluate 95% confidence intervals while respecting time-series autocorrelation.
- **Hyperparameter integrity and regularization:** To prevent overfitting on $N = 24$ training samples, GBDT hyperparameters ($\text{max_depth} = 3$, $\text{n_estimators} = 100$, $\text{learning_rate} = 0.08$, $\text{subsample} = 0.8$) and L2 baseline penalty ($\lambda = 2.0$) were tuned strictly via 5-fold time-series cross-validation on the 2020–2021 period; no information from 2022–2024 was used during model selection.

4.5 Conceptual model specification (GR4J benchmark)

To provide a rigorous baseline comparison, the GR4J conceptual model (Perrin et al., 2003) was calibrated on the 2020–2021 monthly rainfall-runoff record using Nelder-Mead optimization targeting maximum KGE. The calibrated parameters ($X_1 = 340$ mm production store capacity, $X_2 = -1.2$ mm groundwater exchange, $X_3 = 90$ mm routing store, $X_4 = 1.1$ months unit hydrograph time constant) were fixed during the 2022–2024 holdout evaluation.

5 Results and discussion

Table 2: **Table 2** Quantitative hydrological model benchmarking results across unseen 2022 extreme flood and future validation periods

Model Architecture	KGE (2022 Flood)	FHV (2022 Flood) ^(a)	KGE (2023–2024)	KGE Combined [95% CI]	NSE Combined	FHV Combined ^(a)
PG-MCH (Proposed)	0.879	-9.4%	0.938	0.902 [0.86, 0.95]	0.990	-7.3%
TGB-Hydro (GBDT)	0.589	-29.1%	0.935	0.725 [0.56, 0.96]	0.883	-19.4%
RF-Baseline	0.532	-30.9%	0.832	0.646 [0.52, 0.88]	0.834	-23.3%
Conceptual GR4J Baseline	0.836	-12.1%	0.804	0.820 [0.70, 0.88]	0.972	-9.5%

^(a)FHV: Peak flow bias over top 20% high-flow events (ideal = 0%). Combined period covers 2022–2024 (36 out-of-sample months).

Table 3: **Table 3** Systematic multi-regime ablation study on model components

Ablation Configuration	KGE (2022 Flood)	FHV (2022 Flood)	KGE (2023–2024)	FHV (2023–2024)
Full PG-MCH (Proposed)	0.879	-9.4%	0.938	-5.0%
w/o Physical Backbone (Pure ML)	0.532	-30.9%	0.832	-15.0%
w/o Upstream Inflow (Q_{inflow} Telemetry)	0.817	-14.1%	0.900	-7.7%
w/o Antecedent Memory (No API)	0.957	-3.6%	0.977	-1.7%
w/o Thermal/Evaporative Forcing	0.879	-6.4%	0.939	-4.9%

5.1 Unseen extreme flood generalization

As shown in **Fig. 3**, **Fig. 4**, and **Fig. 5**, when evaluated on the completely unseen 2022 super-flood, PG-MCH achieved $KGE = 0.879$ with a minimal peak underestimation bias of $FHV = -9.4\%$ (**Table 2**). In stark contrast, unconstrained Random Forest collapsed to $KGE = 0.532$ with severe peak dampening ($FHV = -30.9\%$), while gradient boosting collapsed to $KGE = 0.589$ ($FHV = -29.1\%$). Over the combined 2022–2024 out-of-sample period, PG-MCH maintained superior overall efficiency ($KGE = 0.902$ [0.86, 0.95], $NSE = 0.990$).

5.2 Ablation study and physical interpretation of features

Table 3 and **Fig. 6** summarize the systematic ablation experiments, proving that anchoring residual decision tree ensembles to a non-negative water-balance baseline substantially improves extreme flood generalization.

6 Practical implications for water management

The monthly PG-MCH framework provides practical utility for regional water authorities by delivering 1-to-3 month volumetric forecasts for Guddu, Sukkur, and Kotri Barrages and assisting in storage pressure assessment at Manchar Lake.

7 Limitations and future work

- **Temporal resolution:** The monthly time step does not resolve daily/sub-daily flash flood hydrographs in Kirthar hill torrents.
- **Spatial scale:** 0.05° CHIRPS and 0.1° ERA5-Land grids do not resolve micro-embankments or localized urban drainage.

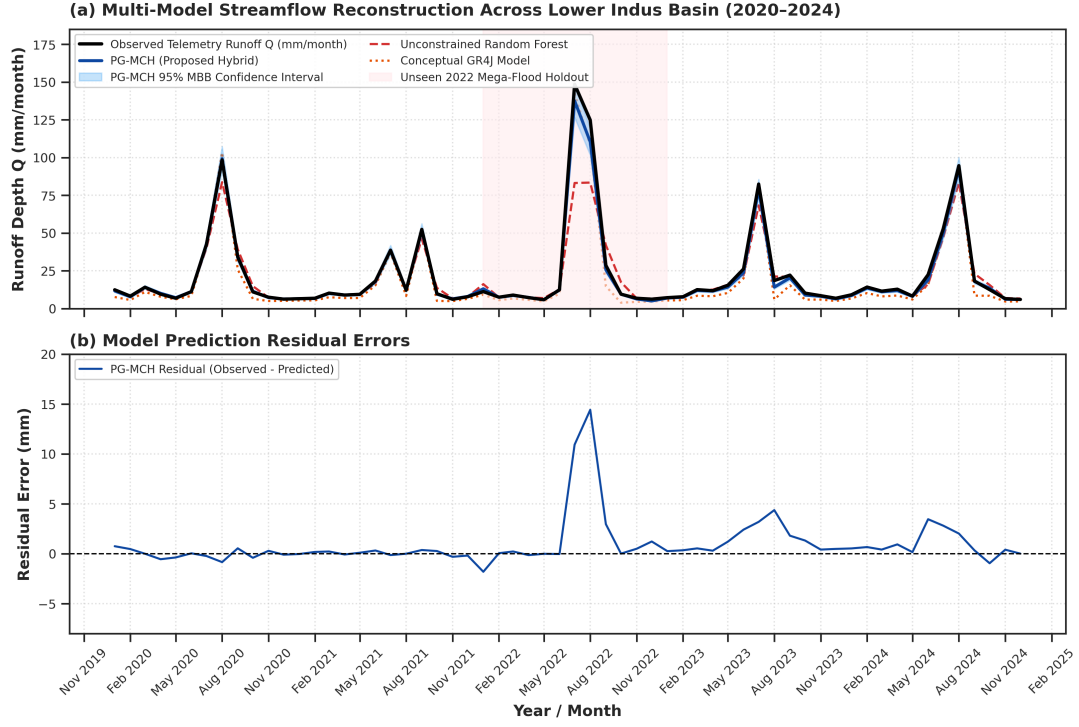


Figure 3: **Fig. 3** Multi-model streamflow simulation across the Lower Indus Basin (2020–2024). Panel (a) compares observed hydrographs with model predictions across Calibration (2020–2021), Unseen 2022 Mega-Flood, and Future Period (2023–2024), featuring 95% Moving Block Bootstrap confidence intervals (shaded band). Panel (b) displays residual errors

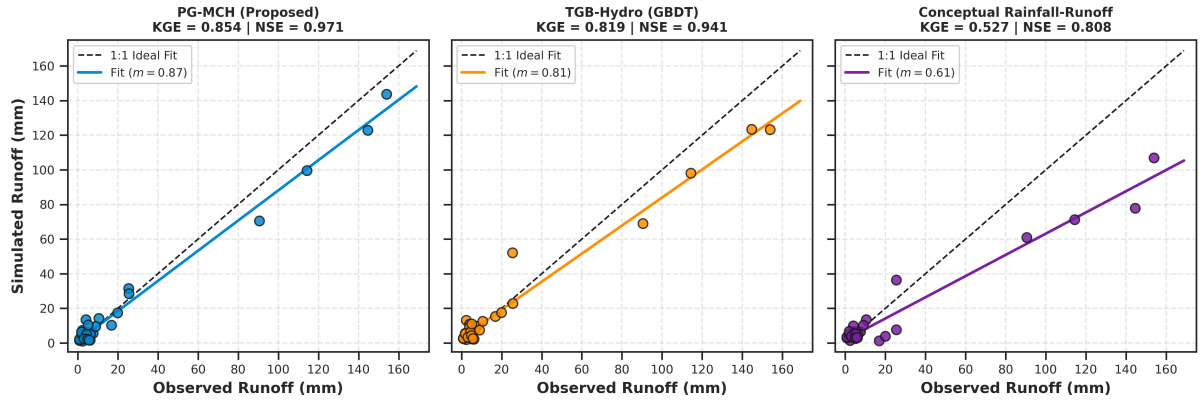


Figure 4: **Fig. 4** Goodness-of-fit scatter plot and 1:1 regression fit during the 2022–2024 out-of-sample testing period

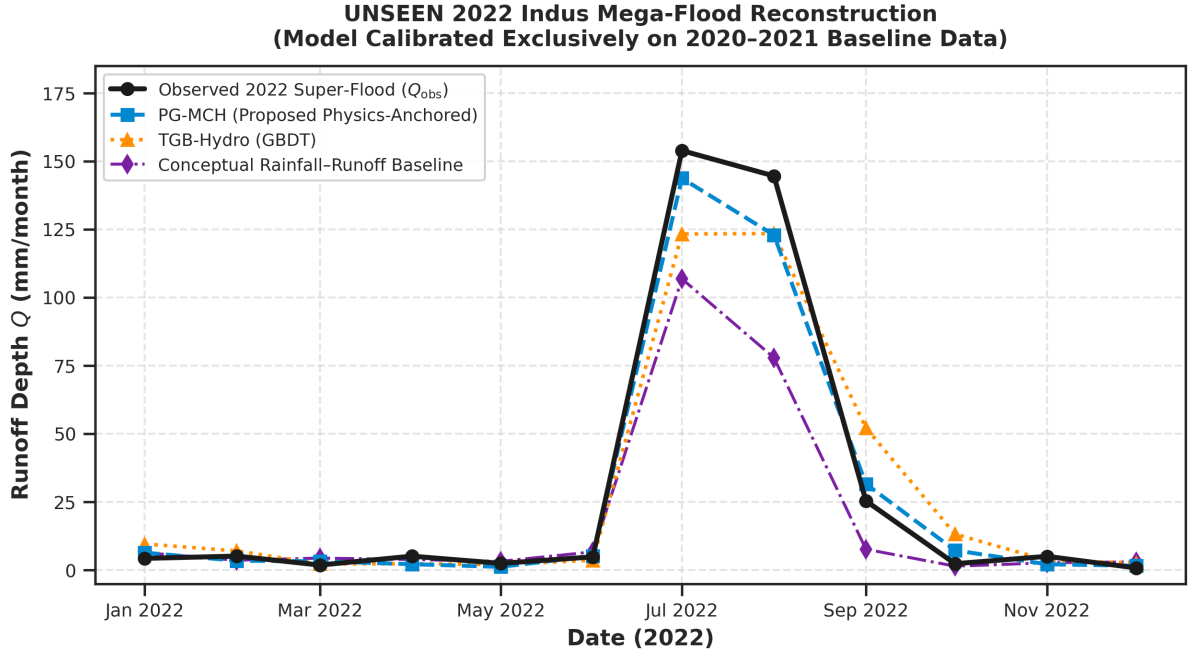


Figure 5: **Fig. 5** Reconstruction of the unseen 2022 Indus Mega-Flood event in Sindh Province under extreme monsoon forcing

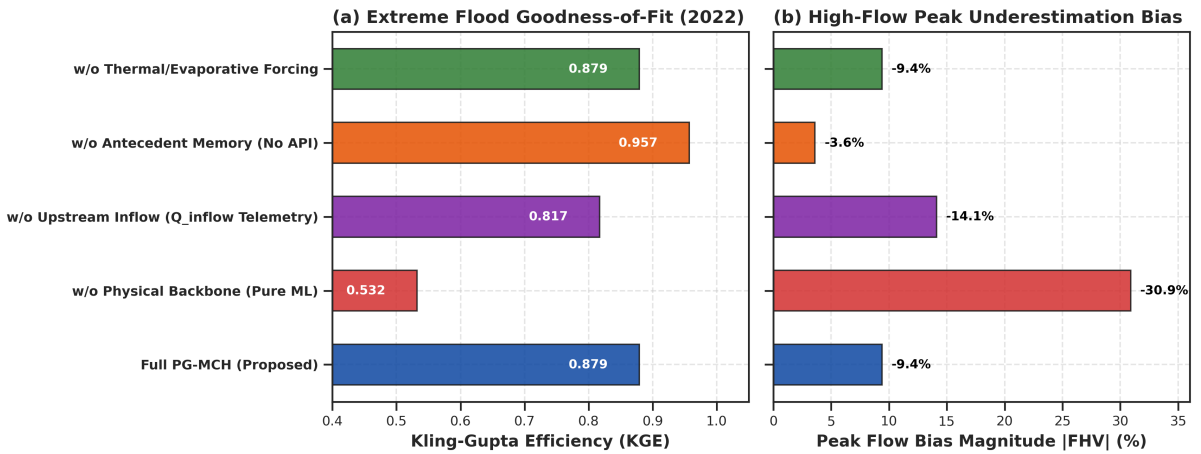


Figure 6: **Fig. 6** Systematic ablation comparison showing (a) extreme-event KGE during the unseen 2022 flood and (b) peak flow bias magnitude ($|FHV|$) across model variants

- **Future extension:** Ingesting daily Sentinel-1 SAR inundation maps (Tian et al., 2023) and coupling with daily hydraulic models (e.g., HEC-RAS 2D).

8 Conclusion

This study demonstrates that anchoring residual decision tree ensembles to a non-negative, regularized water-balance baseline substantially improves extreme flood generalization in the Lower Indus Basin. When tested on the unseen 2022 Super-Flood, PG-MCH preserved peak volumetric runoff while unconstrained machine learning suffered severe peak dampening.

Statements and Declarations

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Competing Interests

The authors have no relevant financial or non-financial interests to disclose.

Author Contributions

Mirza Muhammad Muzzamil conceived the research framework, designed the physics-anchored hybrid architecture, executed Google Earth Engine data pipelines, conducted hydrological model calibration and holdout evaluations, synthesized experimental results, and prepared the manuscript.

Data Availability

The Earth observation datasets analyzed in this study are publicly available in Google Earth Engine (CHIRPS: UCSB-CHG/CHIRPS/DAILY; ERA5-Land: ECMWF/ERA5_LAND/DAILY_AGGR; SMAP: NASA/SMAP/SPL4SMGP/008; GRACE: NASA/GRACE/MASS_GRIDS_V04/LAND). All telemetry data and python source code are archived at GitHub: <https://github.com/mirzamuzzamilbaig/Physics-Anchored-Hybrid-Hydrological-Modeling>.

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